

BOOKING FLOW — FUNCTIONAL DOCUMENTATION

How a meeting room is booked, who gets told, and how clashes are prevented.

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1 PURPOSE

This document describes exactly what happens when someone books a meeting room at the Mumbai branch — the order the questions are asked, what the system does with the answers, who is notified, and how two people are stopped from taking the same room at the same time.

It covers five rooms: **Spark, Power, Pulse, Ignite and Switch**. Developers should build to this document; anything not written here should be raised as a question rather than assumed.

THE ONE RULE THAT MATTERS MOST

If Power is booked from 1 pm to 3 pm, nobody else can see or select that window *for Power*. The same window stays fully available on Spark, Pulse, Ignite and Switch. Availability is always per room, never global.

2 THE BOOKING FLOW

Six steps, in this order. The order matters: the room is chosen first, which is what allows the time picker to grey out the windows that room is already taken for.

1 SELECT THE ROOM

All five rooms are shown. The user picks one. Rooms are shown by name only.

Why first: until we know which room, we cannot know which times are unavailable.

2 SELECT THE DATE

Today up to 90 days ahead. Sundays and declared holidays are not selectable.

3 SET THE ENTRY TIME

A clock-style picker, the same interaction as setting an alarm. Windows already booked for the chosen room on the chosen date are disabled and cannot be selected.

4 SET THE EXIT TIME

The same clock picker, opening at entry time + 30 minutes. It cannot go earlier than the entry time, and it cannot run past the start of the next booking for that room.

5 ENTER THE MEETING DETAILS

Department, who is conducting the meeting, and the attendees.

6 CONFIRM

The system re-checks the room is still free, saves the booking, and sends notifications. The slot immediately disappears from everyone else's view of that room.

BOOKED WINDOWS FOR THIS ROOM ARE DISABLED

1 · ROOM

2 · DATE

3 · ENTRY

4 · EXIT

5 · DETAILS

6 · CONFIRM

DECIDES WHAT
TIMES ARE FREE

RE-CHECK, SAVE,
NOTIFY

3 FIELDS

In the order they appear on screen.

#	FIELD	TYPE	REQUIRED	RULES
1	Room	Choice of five	YES	Spark, Power, Pulse, Ignite, Switch. Name only — no seat counts or equipment shown.
2	Date	Date picker	YES	Today to today + 90 days. Sundays and holidays blocked.
3	Entry time	Clock picker	YES	09:00–19:30, 30-minute steps. Booked windows for the chosen room are disabled.
4	Exit time	Clock picker	YES	Must be after entry. Minimum 30 minutes, maximum 4 hours. Cannot cross into the next booking.
5	Department	Dropdown	YES	Defaults to the booker's own department, editable.
6	Conducted by	Person picker	YES	Who is running the meeting. Defaults to the person booking, but can be someone else.
7	Attendees	Multi-select	NO	Colleagues from the company directory. Each one gets a notification.
8	Meeting title	Text	NO	Shown on the grid. Defaults to "Meeting" if left blank.
9	Note for reception	Text area	NO	Projector, tea, seating. Appears on the front desk sheet.

REMOVED FROM THE EARLIER DESIGN

GOOGLE MEET LINK

— no video-conference link is created. NM Meet books a physical room only.

"TELL RECEPTION" TOGGLE

— removed because reception is *always* notified. It is not optional, so it should not be presented as a choice.

4 TIME SELECTION

Entry and exit times are set with a clock-face picker, the same interaction people already know from setting an alarm on their phone — tap the hour on the dial, then the minutes, then AM or PM.

BEHAVIOUR

- **Minutes snap to 30.** Only :00 and :30 can be selected. A free dial would let someone pick 1:07 pm, which cannot be reconciled with 30-minute slots.
- **Disabled ranges are visible on the dial.** Hours already booked for the chosen room are drawn greyed out and cannot be tapped, so the user sees the constraint rather than being told after the fact.
- **Exit opens at entry + 30 minutes** and cannot be dragged earlier.
- **Exit stops at the next booking.** If Power is free from 12:00 but booked from 13:00, an entry of 12:30 allows an exit of 13:00 only.
- **Outside 09:00–20:00 is disabled** on both dials.
- **Keyboard and typed entry must also work.** A dial alone fails accessibility and is slow on a desktop with a keyboard — allow typing `14:30` directly into the field.

WHY NOT A PLAIN DROPDOWN

A dropdown of 22 times is faster on a desktop but poor on a phone, which is where most bookings will be made. The clock picker is the phone-first choice; the typed field is the desktop escape hatch. Both write to the same value.

5 AVAILABILITY RULES

THE CORE RULE

Two bookings clash when **all three** are true:

- 1. Same room
- 2. Same date
- 3. The times overlap

Times overlap when:

```
new.entry < existing.exit AND new.exit > existing.entry
```

Note the strict `<` and `>`. A meeting ending at 15:00 and another starting at 15:00 do **not** clash — they are back to back, which is allowed.

WORKED EXAMPLE

Rahul books **Power, 1 pm to 3 pm** on 26 August.

SOMEBODY ELSE TRIES	RESULT	WHY
Power · 1:00–3:00 pm	Blocked	Exactly the same window
Power · 2:00–4:00 pm	Blocked	Overlaps 2:00–3:00
Power · 12:00–2:00 pm	Blocked	Overlaps 1:00–2:00
Power · 12:00–1:00 pm	Allowed	Ends exactly when the other starts
Power · 3:00–4:00 pm	Allowed	Starts exactly when the other ends
Pulse · 1:00–3:00 pm	Allowed	Different room — Power's booking is irrelevant
Switch · 1:00–3:00 pm	Allowed	Different room



WHAT THE USER ACTUALLY SEES

- After choosing Power and 26 August, the 1 pm to 3 pm range is **greyed out and struck through** on the entry dial.
- If they pick 12:30 as entry, the exit dial offers **13:00 only**.

- Choosing Pulse instead immediately re-enables the full day.
- If every room is taken for a window, the message reads: "All five rooms are booked between 1 pm and 3 pm. The first free room is Pulse at 3 pm."

6 TRY IT

A working demonstration of the rule above. Power is booked 1 pm to 3 pm and Ignite 10 am to 11 am. Pick a room, then click a start time and an end time.

STEP 1 – PICK A ROOM

Spark

Power

Pulse

Ignite

Switch

STEP 2 – PICK ENTRY, THEN EXIT

09:00

09:30

10:00

10:30

11:00

11:30

12:00

12:30

13:00

13:30

14:00

14:30

15:00

15:30

16:00

16:30

17:00

17:30

18:00

18:30

19:00

19:30

POWER — already booked 1 pm–3 pm. Those times are struck through and cannot be picked.
Every other room stays fully available in this window.

7 PREVENTING CLASHES

Hiding booked times in the interface is not enough. Two people can press Confirm within the same second, and both will have been shown the slot as free.

Three layers are required, and all three must be built:

LAYER	WHAT IT DOES	WHY IT IS NOT SUFFICIENT ALONE
1 • Screen	Greys out booked windows for the chosen room	The data can go stale between opening the form and pressing Confirm
2 • Server	Re-checks availability at the moment of saving	Two requests can pass this check simultaneously before either has written
3 • Database	Physically refuses to store an overlapping row	This is the only guarantee

Layer 3 in PostgreSQL:

```
CREATE EXTENSION IF NOT EXISTS btree_gist;

ALTER TABLE bookings
ADD CONSTRAINT no_double_booking
EXCLUDE USING gist (
    room_id WITH =,
    tstzrange(entry_time, exit_time, '[') WITH &&
) WHERE (status = 'CONFIRMED');
```

The '[' means the range includes its start and excludes its end — which is precisely why a booking ending at 15:00 does not clash with one starting at 15:00.

WHAT THE LOSER SEES

The second person gets a clear message rather than a database error: *"Power was booked by someone else a moment ago. Pulse and Switch are still free from 1 pm to 3 pm."* The screen refreshes with the correct availability.

8 NOTIFICATIONS

Notifications are **always sent**. There is no toggle and no opt-out.

RECIPIENT	WHO THAT IS	BOOKED	CHANGED	CANCELLED
Attendees	Everyone added in the attendees field	Yes	Yes	Yes
Conducting	The person running the meeting	Yes	Yes	Yes
Reception	The Mumbai front desk mailbox	Yes	Yes	Yes
Mumbai group	The branch-wide distribution list	Yes	Yes	Yes
Booker	Whoever filled in the form	Yes	Yes	Yes

WHAT THE MESSAGE CONTAINS

- Room, date, entry and exit time
- Meeting title and department
- Who is conducting it
- The attendee list
- The note for reception, if one was given
- A link to view or cancel the booking

WORTH CONFIRMING

Emailing the whole Mumbai branch on every single booking may become noise very quickly. Consider sending the branch group a

DAILY 8 AM SUMMARY

instead of a message per booking, while attendees, the person conducting and reception continue to be notified immediately. This is a decision for the business, not the developers.

9 CANCELLING

Every booking has a **Cancel** button on its detail view.

WHO CAN CANCEL

ROLE	CAN CANCEL
The person who booked it	Their own bookings
The person conducting it	That meeting
Reception	Any booking
Admin	Any booking
An attendee	No — they can only decline

WHAT HAPPENS

- 1. The button names the owner, for example "Cancel Rahul's booking", so nobody cancels the wrong meeting.
- 2. The first click asks for confirmation. The button turns red and reads "Click again to confirm".
- 3. On the second click the booking is cancelled.
- 4. The room becomes selectable again for that window, immediately, for everyone.
- 5. A cancellation notice goes to all five recipient groups in section 8.
- 6. The booking is marked CANCELLED and kept for the audit record — it is never deleted.

CANCELLING AFTER THE MEETING HAS STARTED

Allowed, and it releases the remaining time. A 1 pm to 3 pm booking cancelled at 1:40 pm frees the room from 1:40 pm onward for everyone else.

10 VALIDATION AND MESSAGES

SITUATION	MESSAGE SHOWN
Exit time is before or equal to entry time	The meeting has to end after it starts.
Shorter than 30 minutes	The shortest booking is 30 minutes.
Longer than 4 hours	Four hours is the longest single booking. Split it into two.
Outside 09:00–20:00	Rooms can be booked between 9 am and 8 pm.
Date is in the past	That date has passed.
Date is a Sunday or a holiday	The office is closed that day.
Taken while the form was open	Power was booked by someone else a moment ago. Pulse and Switch are still free from 1 pm to 3 pm.
No room free for the chosen window	All five rooms are booked between 1 pm and 3 pm. The first free room is Pulse at 3 pm.
Department not chosen	Please choose a department.
Nobody named as conducting	Please say who is conducting the meeting.

Every message names the room and the time. "Invalid selection" or "Booking failed" is not acceptable — the user must be told what to do next.

11 WHAT IS STORED

```

bookings
  id
  room_id      → spark | power | pulse | ignite | switch
  booking_date  → date
  entry_time    → timestamptz, stored UTC, shown Asia/Kolkata
  exit_time     → timestamptz
  department_id
  conducted_by  → user id
  booked_by     → user id (may differ from conducted_by)
  title         → text, defaults to "Meeting"
  reception_note → text, nullable
  status        → CONFIRMED | CANCELLED | NO_SHOW
  created_at, updated_at

booking_attendees
  booking_id, user_id, response_status

notification_log
  booking_id, recipient, event (BOOKED|CHANGED|CANCELLED), sent_at, status

audit_log
  entity, entity_id, action, actor_id, before, after, created_at

```

Store times as `timestamptz` in UTC and display in Asia/Kolkata. Never store `"13:00"` as text — it breaks the moment anything crosses a date boundary or the server moves.

12 TEST CASES

Hand these to QA as written. Each must pass before sign-off.

#	TEST	EXPECTED
T-01	Book Power 1–3 pm. Open a new booking, choose Power, same date.	1 pm to 3 pm is disabled on the entry dial.
T-02	Same as above, but choose Pulse.	The whole day is selectable.
T-03	Power booked 1–3 pm. Book Power 12–1 pm.	Succeeds — back to back is allowed.
T-04	Power booked 1–3 pm. Book Power 3–4 pm.	Succeeds.
T-05	Power booked 1–3 pm. Choose Power, entry 12:30.	Exit offers 1 pm only.
T-06	Two people confirm the same room and window simultaneously.	One succeeds. The other sees a named message and a refreshed screen.
T-07	Try to set the exit before the entry.	Rejected with a plain message.
T-08	Try to book 7 pm to 9 pm.	Rejected — outside working hours.
T-09	Complete a booking with two attendees.	Five notifications sent: two attendees, conductor, reception, Mumbai group.
T-10	Cancel that booking.	The window frees immediately for everyone; cancellation notices sent.

T-11	Reception cancels someone else's booking.	Allowed, after the two-click confirm.
T-12	An attendee tries to cancel.	No cancel button is shown.
T-13	Cancel a booking that is already running.	Remaining time is released.
T-14	Book all five rooms 1–3 pm, then try a sixth booking.	Message names the first room to free up.
T-15	Set the minutes on the clock picker to :15.	Not possible — minutes snap to :00 and :30.

13 OPEN DECISIONS

#	QUESTION	RECOMMENDATION
D-01	Should the Mumbai group get every booking, or a daily summary?	Daily 8 am summary — per-booking email will become noise
D-02	Should a calendar invite still be attached, now that Google Meet is dropped?	Yes — the invite is how it reaches people's calendars; only the video link is removed
D-03	Working days — is Saturday included? Is Sunday always closed?	Monday to Saturday, Sunday closed
D-04	Can the person conducting be someone outside the company?	No for now — directory members only
D-05	Is a gap needed between two meetings in the same room?	No — back to back allowed, as documented
D-06	How long before an unused room is released as a no-show?	15 minutes, released by reception
D-07	Minute granularity — 30 minutes or 15?	30 minutes